

Individual Assignment 5

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1. Set Up

Packages

```
# general use
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic info
taxon_list <- read_csv(here("data","taxon_list.csv"))
```

```

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects"
)

# site info
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

2. Cleaning

sites object

```

# creating new object
ncos_sites <- sites |>
  # filtering to only include ncos quarterly sampling sites
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

clean taxon_list

```

taxon_list_clean <- taxon_list |>
  # convert taxon names to snakecase
  mutate(verbatim_name = to_any_case(verbatim_name,
    case = "snake"))

```

clean water_quality

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = FALSE) |>
  group_by(date, site, date_site) |>
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  ungroup() |>
  filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

```
# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
    annelida_polychaete) |>
  # filter to only include "filtered beaker" samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # convert the data frame to long format
  pivot_longer(
    cols = ostracod:annelida_polychaete,
    names_to = "taxon",
    values_to = "abundance") |>
```

```

# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↳ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(
  select(water_quality_clean,
        date_site,
        med_ph,
        med_sal,
        med_do),
  by = "date_site"
) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↳ = TRUE),
  site_full = fct_rev(site_full))

```

clean aq_ins_clean

```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>

```

```
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))
```

Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site
- y-axis: ostracod abundance
- geometry to represent average abundance/L: `geom_jitter`
- geometry to represent medians: `stat_summary()`

b. Wrangle your data

```
# create new object from aq_ins_clean
ost_abundance <- aq_ins_clean |>
# filter to only ostracods
filter(taxon == "ostracod") |>
# remove missing abundance/L values
filter(!is.na(ave_lit)) |>
# group by site
group_by(site_full) |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>
# ungroup dataframe
ungroup()

slice_sample(ost_abundance, n = 10)
```

A tibble: 10 x 24

	date_site	date	site	taxon	ave_lit	med_ph	med_sal	med_do
	<chr>	<dtm>	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13	NA	NA	NA
2	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	ostr~	2	7.68	2.36	2.43

```

3 2022-01-28_NDC 2022-01-28 00:00:00 NDC ostr~ 9.47 7.65 2.00 9.55
4 2023-09-15_NMC 2023-09-15 00:00:00 NMC ostr~ 0.4 NA NA 14.2
5 2023-02-21_NEC 2023-02-21 00:00:00 NEC ostr~ 0 8.56 NA 8.8
6 2023-11-14_NPB 2023-11-14 00:00:00 NPB ostr~ 18 NA NA NA
7 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1 ostr~ 0.533 7.48 NA 2.38
8 2023-08-12_NEC 2023-08-12 00:00:00 NEC ostr~ 0 NA NA NA
9 2023-02-28_NPB1 2023-02-28 00:00:00 NPB1 ostr~ 0 NA NA NA
10 2022-02-25_NEC 2022-02-25 00:00:00 NEC ostr~ 0 8.75 44.1 9.87
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
# Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
# Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
# comments <chr>, site_full <fct>, log_ave_lit <dbl>

```

c. Code your figure

```

# base layer
ostracods_plot <- ggplot(data = ost_abundance,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
  fun = "median",
  size = 4,
  color = "darkgreen") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Ostracod Abundance") +
# different theme
theme_bw()

# displaying object
ost_abundance

```

```
# A tibble: 51 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4	6.53	0.3	7.6
2	2020-02-21_NVBR	2020-02-21 00:00:00	NVBR	ostr~	4.8	7.34	4.1	6.1
3	2021-04-15_NPB	2021-04-15 00:00:00	NPB	ostr~	185.	8.09	3.57	8.24
4	2021-08-25_NPB	2021-08-25 00:00:00	NPB	ostr~	581.	7.52	4.57	4.5
5	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	ostr~	0	NA	NA	NA
6	2021-11-12_NPB	2021-11-12 00:00:00	NPB	ostr~	79.2	NA	NA	NA
7	2021-11-16_NPB	2021-11-16 00:00:00	NPB	ostr~	72.1	7.3	2.38	12.1
8	2021-11-23_NPB2	2021-11-23 00:00:00	NPB2	ostr~	0.267	7.59	1.99	0.65
9	2022-01-21_NEC	2022-01-21 00:00:00	NEC	ostr~	0	NA	NA	NA
10	2022-01-24_NEC	2022-01-24 00:00:00	NEC	ostr~	0.267	NA	NA	NA

```
# i 41 more rows
```

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,  
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,  
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,  
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

d. Create a multipanel figure

```
# base layer  
salinity_plot <- ggplot(data = aq_ins_clean,  
  # x-axis: site  
  # y-axis: mean salinity  
  mapping = aes(x = site_full,  
    y = med_sal)) +  
  # first layer: salinity measurements at each survey  
  geom_jitter(height = 0,  
    width = 0.1,  
    shape = 21) +  
  # wrapping x-axis text  
  scale_x_discrete(labels = label_wrap(width = 15)) +  
  # second layer: points to represent means  
  stat_summary(fun = median,  
    color = "red",  
    size = 1) +  
  # relabelling x- and y-axes  
  labs(x = "Site",  
    y = "Median salinity (ppt)",  
    title = "Site Salinity") +
```

```

# different theme
theme_bw()

# base layer
copepods_plot <- ggplot(data = copepods,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "Average abundance/L",
     title = "Copepod Abundance") +
# different theme
theme_bw()

# base layer
ostracods_plot <- ggplot(data = ost_abundance,
                        # x-axis: site
                        # y-axis: mean abundance/L
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "darkgreen") +
# wrapping x-axis text

```



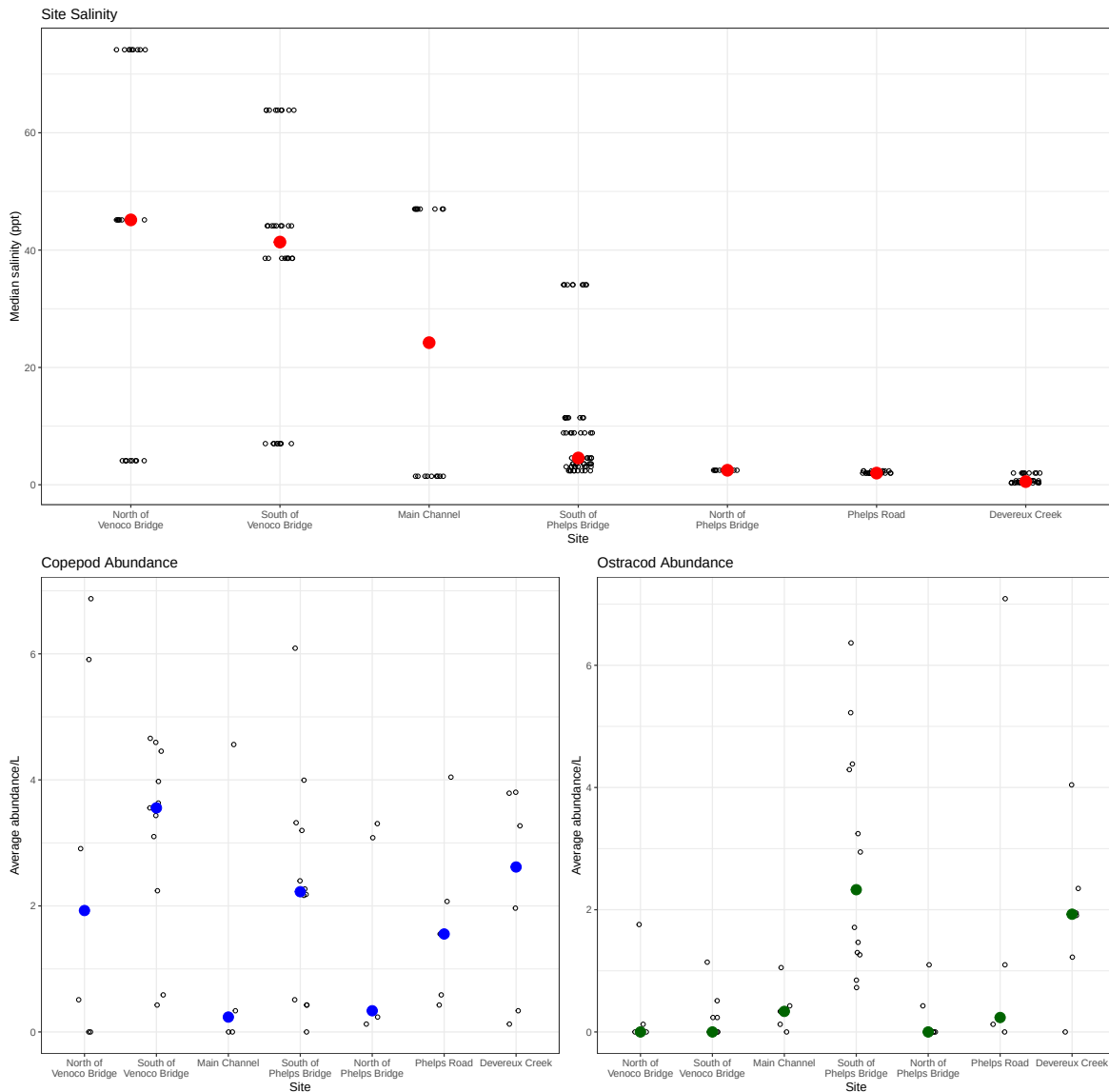
```

scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
      y = "Average abundance/L",
      title = "Ostracod Abundance") +
# different theme
theme_bw()

(salinity_plot) / (copepods_plot + ostracods_plot) +
plot_annotation(title = "Comparing Salinity, Copepod Abundance, and
  ↪ Ostracod Abundance across sites at NCOS")

```

Comparing Salinity, Copepod Abundance, and Ostracod Abundance across sites at NCOS



e. Interpret your figure

differences between sites in copepod and ostracod abundance how ostracod abundance may relate (or not) to site-level differences in salinity

2. Visualizing total abundance as a function of salinity

a. Plan your figure

x-axis: y-axis: color: geometry: facets:

b. Wrangle your data

c. Code your figure

d. Interpret your figure

In 3-5 sentences, describe differences or similarities between taxa in relationships between abundance and salinity. Be specific using the visual components of the plot.